Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The mass of $BaCO_3$ produced when excess of CO_2 is bubbled through a solution of $0.205\ mol Ba(OH)_2$ is

(A) $81\ g$

(B) $40.5\ g$

(C) $20.25\ g$

(D) $162\ g$

Q.2

The ratio of amounts of $H_2\ S$ needed to precipitate all the metal ions from $100ml$ of $1M$ $AgNO_3$ and $100ml$ of $1M CuSO_4$ will be

(A) $1:1$

(B) $1:2$

(C) $2:1$

(D) None of these

Q.3

$0.45\ g$ of acid (molecular weight 90) is neutralised by $20ml$ of $0.5\ N$ caustic potash. The basicity of acid is

(A) 1

(B) 2

(C) 3

(D) 4

Q.4

$0.75\ g$ platinic chloride of a mono- acidic base on ignition gives $0.245\ g$ platinum. The molecular weight of the base is molecular weight of $Pt = 195$

(A) 75.0

(B) 93.5

(C) 100

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(D) 80.0

Q.5

Given :

$XNa_2HAsO_3 + YNaBrO_3 + ZHCl \rightarrow$
$NaBr + H_3AsO_4 + NaCl$

The values of X, Y and Z in the above redox reaction are respectively :

(A) $2, 1, 2$

(B) $2, 1, 3$

(C) $3, 1, 6$

(D) $3, 1, 4$

Q.6

The number of electrons involved in the reduction of one nitrate ion to hydrazine is

(A) 8

(B) 5

(C) 3

(D) 7

Q.7

In the following balanced reaction,

$XMnO_4^- + YC_2O_4^{2-} + ZH^+$
$\rightleftharpoons XMn^{2+} + 2YCO_2 + \frac{Z}{2}H_2O$

values of X, Y and Z respectively are

(A) $2, 5, 16$

(B) $8, 2, 5$

(C) $5, 2, 16$

(D) $5, 8, 4$

Q.8

1 mole of equimolar mixture of $Fe_2(C_2O_4)_3$ and FeC_2O_4 requires X moles of $KMnO_4$ in acid medium for complete reaciton. The value of X is:

(A) 0.9

(B) 0.6

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(C) 1.2

(D) 0.8

Q.9

For the reaction:
$NH_3 + OCl^- \longrightarrow N_2H_4 + Cl^-$ in basic
medium, the coefficients of NH_3, OCl^- and N_2H_4 for the balanced equation are respectively

(A) $2, 2, 2$

(B) $2, 2, 1$

(C) $2, 1, 1$

(D) $4, 4, 2$

Q.10

Zn gives H_2 gas with H_2SO_4 and HCl but not with HNO_3 because

(A) Zn acts as an oxidising agent when it reacts with HNO_3

(B) HNO_3 is weaker acid than H_2SO_4 and HCl

(C) In electrochemical series, Zn is above hydrogen

(D) NO_3^- is reduced in preference to hydronium ion

Q.11

One mole of N_2H_4 loses 10 moles of electrons to form a new compound, y. Assuming that all nitrogen appear in the new compound, what is the oxidation state of nitrogen in y (There is no change in the oxidation state of hydrogen)

(A) -1

(B) -3

(C) $+3$

(D) $+5$

Q.12

In the reaction $3Mg + N_2 \rightarrow Mg_3\ N_2$

(A) magnesium is reduced

(B) magnesium is oxidized

(C) nitrogen is oxidized

(D) None of these

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Q.13

Which of the following involves transfer of five electrons ?

(A) $MnO_4^- \rightarrow Mn^{2+}$

(B) $CrO_4^{2-} \rightarrow Cr^{3+}$

(C) $MnO_4^{2-} \rightarrow MnO_2$

(D) $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$

Q.14

The colour of phenolphthalein in acid and base ranges of the indicator respectively, are

(A) pink, colourless

(B) pink, red

(C) colourless, pink

(D) red, pink

Q.15

Which of the following indicators can be used to detect end point in the titration of a weak acid versus strong base?

(A) Methyl red $(4.2$ to $6.3)$

(B) Methyl violet $(0.1$ to $3.2)$

(C) Methyl orange $(3.1$ to $4.4)$

(D) Thymolphthalein $(9.3$ to $10.5)$

Q.16

The titration of a weak base versus weak acid cannot be carried out by using acid-base indicator because

(A) there is no variation in pH of the solution

(B) the variation in pH is gradual with no steep change in pH near the equivalence point.

(C) the change in pH near the equivalence point does not encompasses an interval equal to the pH transition range of the indicator

(D) no indicator can be found which changes colour at pH of equivalence point.

Q.17

Which of the following indicators (buffer range provided in brackets) cannot be used to detect end point of titration between strong acid and strong base?

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(A) Trinitrobezene $(12.0 \text{ to } 14.0)$

(B) Phenol red $(6.0 \text{ to } 7.6)$

(C) p-Nitrophenol $(5.0 \text{ to } 7.0)$

(D) Phenolphthalein $(8.3 \text{ to } 10.0)$

Q.18

The colour of methyl orange in acid and base ranges of the indicator, respectively, are

(A) yellow, red

(B) red, yellow

(C) yellow, blue

(D) blue, red

Q.19

Find the value of 'b' for the following half equation.

$Mn^{2+} + xH_2O \longrightarrow MnO_2 + aH^+ + be^-$

(A) 2

(B) 1

(C) 4

(D) 3

Q.20

The product of oxidation of I^- with MnO_4^- in alkaline medium is:-

(A) IO_3^-

(B) I_2

(C) IO^-

(D) IO_4^-

Answers & Solutions

Q.1 Answer:

40.5 g

Solution:

$Ba(OH)_2 + CO_2 \rightarrow BaCO_3 + H_2O$

Atomic wt. of $BaCO_3 = 137 + 12 + 16 \times 3 = 197$

No. of mole $= \frac{wt \text{ of substance}}{\text{mol wt.}}$

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$\because$ **1 mole of** $\mathrm{Ba(OH)_2}$ **gives 1 mole of** $\mathrm{BaCO_3}$

$\therefore$ **205 mole of** $Ba(OH)_2$ **will give 205 mole of** $BaCO_3$

$\therefore$ **wt. of** 0.205 **mole of** $\mathrm{BaCO_3}$ **will be**

$0.205 \times 197 = 40.385\mathrm{gm} = 40.5\mathrm{gm}$

Q.2 Answer:

$1 : 2$

Solution:

$\mathrm{AgNO_3} \equiv 2\mathrm{Ag^+} + \underset{(H_2\ \mathrm{S})}{\mathrm{S^{2-}}} \rightarrow \mathrm{Ag_2\ S}$

$\because$ **2 mole** $\rightarrow$ **1 mole** $\quad [100 \times 1 = 100 millimole]$

$\therefore$ 100 **millimole** $\rightarrow$ 50 **millimole** H_2S **required**

$\mathrm{CuSO_4} \equiv \mathrm{Cu^{2+}} + \underset{(H_2\ \mathrm{S})}{\mathrm{S^{2-}}} \rightarrow \mathrm{Cu\ S}$

1 mole $\rightarrow$ **1 mole** $(100 \times 1 = 100$ **millimole** $)$

100 milimole $\rightarrow$ 100 **milimole** $\mathrm{H_2\ S}$ **required**

Ratio $\frac{50}{100} = \frac{1}{2}$

Q.3 Answer:

2

Solution:

Eq. of acid = Eq of base,

$\therefore \frac{0.45}{\mathrm{E.wt}} = \frac{20 \times 0.5}{1000}, = \mathbf{E.}\ \mathrm{wt} = 45$

Basicity $= \frac{\mathrm{M.wt}}{\mathrm{E.wt}} = \frac{90}{45} = 2$

Q.4 Answer:

93.5

Solution:

Apply the formula M.wt

$= \frac{\mathrm{Acidity}}{2}\left(\frac{wt\ \mathrm{of\ Pt\ salt} \times 195}{\mathrm{wt\ of\ Pt}} - 410\right)$

$\therefore \mathrm{M.wt} = \frac{1}{2}\left(\frac{0.75 \times 195}{0.245} - 410\right) = 93.5$

Q.5 Answer:

$3, 1, 6$

Solution:

On balancing the given reaction, we find

$$3\mathrm{Na_2HAsO_3} + \mathrm{NaBrO_3} + 6\mathrm{HCl} \longrightarrow 6\mathrm{NaCl} + 3\mathrm{H_3AsO_4} + \mathrm{NaBr}$$

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Q.6 Answer:

7

Solution:

$$\overset{+5}{N}O_3^- \longrightarrow \overset{-2}{N_2}H_4$$

So, for reduction of 1 mole of NO_3^- number of electrons required is 7 .

Q.7 Answer:

$2, 5, 16$

Solution:

$XMnO_4^- + YC_2O_4^{2-} + ZH^+ \leftrightharpoons XMn^{+2} + 2YCO_2 + \frac{Z}{2}H_2O$

First half reaction $MnO_4^- \longrightarrow Mn^{+2}$ **...(i)**

On balancing

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{+2} + 4H_2O$**...(ii)**

Second half reaction

$C_2O_4^{2-} \longrightarrow 2CO_2$**...(iii)**

On balancing

$C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^-$**...(iv)**

...

On multiplying eqn. (ii) by 5 and (iv) by 2 and then adding we get

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \longrightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$$

Q.8 Answer:

0.9

Solution:

$Fe_2C_2O_{43} \longrightarrow 2Fe^{3+} + 6CO_2 + 6e$ n Factor $= 6$

$FeC_2O_4 \longrightarrow Fe^{3+} + 2CO_2 + 3e$ n Factor $= 3$

Meqs of $Fe_2(C_2O_4)_3 + Fe(C_2O_4) = 0.5 \times 6 + 0.5 \times 3$

$= 4.5$ Eqs.

$= 4.5$ Eqs.

of $KMnO_4$

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$= \frac{4.5}{5} = 0.9$

moles of $KMnO_4$.

Q.9 Answer:

$2, 1, 1$

Solution:

The balanced equation :

$2NH_3 + OCl^- \longrightarrow N_2H_4 + Cl^- + H_2O$

Q.10 Answer:

NO_3^- is reduced in preference to hydronium ion

Solution:

Zinc gives H_2 **gas with dil** H_2SO_4/HCl **but not with** HNO_3 **because in** HNO_3, NO_3^- **ion is reduced and give** NH_4NO_3, N_2O, NO **and** NO_2 **(based upon the concentration of** HNO_3 **)**

$[Zn + 2HNO_3 \longrightarrow Zn(NO_3)_2 + 2H] \times 4$

(nearly 6% **)**

$HNO_3 + 8H \longrightarrow NH_3 + 3H_2O$

$NH_3 + HNO_3 \longrightarrow NH_4NO_3$

$4Zn + 10HNO_3 \longrightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$

Zn **is on the top position of hydrogen in electrochemical series. So Zn displaces** H_2 **from dilute** H_2SO_4 **and** HCl **with liberation of** H_2. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

Q.11 Answer:

$+3$

Solution:

$$\overset{-4}{N_2}\overset{+4}{H_4} \xrightarrow[N]{\text{loss of } 10e^-} \overset{+6}{N_2Y};$$

O.N. of N changes from -2 to $+3$

Q.12 Answer:

magnesium is oxidized

Solution:

In the given reaction, oxidation state of Mg **is changing from 0 to** $+2$**, while in nitrogen it is**

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changing from 0 to -3. So oxidation of Mg and reduction of nitrogen takes place.

Q.13 Answer:

$\mathrm{MnO}_4^- \rightarrow \mathrm{Mn}^{2+}$

Solution:

O.N. of Mn in MnO_4^- is $+7$ and in Mn^{2+} it is $+2$ The difference is of 5 electrons

Q.14 Answer:

colourless, pink

Solution:

Phenolphthalein is colourless in acid range and pink in base range of the indicator.

Q.15 Answer:

Thymolphthalein $(9.3$ to $10.5)$

Solution:

In the titration of a weak acid (say, acetic acid) versus strong base (say, sodium hydroxide), the pH at the equivalence point is more that 7 due to hydrolysis of congugate base of the weak acid. The steep rise in pH around equivalence point encompasses the range 7 to 10 . Thus, thymolphthalein can be used as the indicator to locate end point.

Q.16 Answer:

the change in pH near the equivalence point does not encompasses an interval equal to the pH transition range of the indicator

Solution:

During the titration, a buffer solution of weak acid and salt of its cojugate base is formed. Its pH is given by

$\mathrm{pH}=\mathrm{p}K_{\mathrm{a}}^{\circ}+\log([\text{salt}]/[\text{ acid }])$

From the given date, we find

$4.14=\mathrm{p}K_{\mathrm{a}}^{\circ}+\log[10/(50-10)]$

This gives $\mathrm{p}K_{\mathrm{a}}^{\circ}=4.14-\log(1/4)=4.14+0.60=4.74$

After the addition of $40\ \mathrm{mL}$ of NaOH solution, we will have

$\mathrm{pH}=4.74+\log[40/(50-40)]=4.74+0.60=5.34$

Q.17 Answer:

Trinitrobezene $(12.0$ to $14.0)$

Solution:

Indicator range of trinitrobenzene is beyond the steep portion of pH near the equivalence point.

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Q.18 Answer:

red, yellow

Solution:

Methyl orange is red in acid range and yellow in base range of the indicator.

Q.19 Answer:

2

Solution:

Balanced half equation is

$Mn^{2+} + 2H_2O \longrightarrow MnO_2 + 4H^+ + 2e^-$

Value of $'b'$ **is** 2**.**

Q.20 Answer:

IO_3^-

Solution:

Oxidaiton of I^- **with** $KMnO_4$ **in alkaline solution is given by**

Reaction: $6MnO_4^- + I^- + 6OH^- \rightarrow IO_3^- + 6MnO_4^{2-} + 3H_2O$

Here the oxidation state of Mn **reduced from +7 to +6 and the oxidation state of I oxidised from -1 to +5.**

$6e^-$ **is involved in the reaction.**

Oxidation product form is $1O_3^-$**.**

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